

1

PROBLEM

Let ABC be a triangle with $AB = 26$, $BC = 17$, $AC = 25$. Let Ω and H be the circumcircle and orthocentre. Let D, E, F be the feet of the altitudes from A, B, C respectively (so D lies on BC , E on CA , F on AB). Let lines EF and BC meet at X , and let line AX meet Ω again at Q ($Q \neq A$). The segment HQ meets the circle DEF (the nine-point circle) at a point P . If $PQ/PE = a/b$ in lowest terms, find $a + b$.

ANSWER: $PQ/PE = 125/91$, so $a + b = 125 + 91 = 216$.

2

SET-UP: COORDINATES

Place $B = (0, 0)$, $C = (17, 0)$, $A = (10, 24)$.

Check of the side lengths: $BC = 17$, $AB = \sqrt{10^2 + 24^2} = \sqrt{100 + 576} = \sqrt{676} = 26$, $AC = \sqrt{(10-17)^2 + 24^2} = \sqrt{49 + 576} = \sqrt{625} = 25$.

So this triangle is congruent to the given one, and all metric quantities (lengths, ratios) computed below are the required ones.

2.1 Orthocentre H .

Since BC lies on the x -axis, the altitude from A is the vertical line $x = 10$. The line AC has slope $(0-24)/(17-10) = -24/7$, so the altitude from B (through the origin, perpendicular to AC) has slope $7/24$, i.e. $y = (7/24)x$. At $x = 10$ this gives $y = 70/24 = 35/12$. Hence

$$H = (10, 35/12).$$

(Consistency check with the altitude from C : line AB has slope $24/10 = 12/5$, so the altitude from C is $12x - 5y = 170 - \dots$; explicitly $5x + 12y = 85$, which at $x = 10$ gives $y = 35/12$. Agreed.)

2.2 Feet of the altitudes D, E, F .

D = foot from A on $BC = (10, 0)$.

E = foot from B on line AC . Line AC passes through $A(10,24)$ with slope $-24/7$, i.e. $24x + 7y = 408$ (indeed $24 \cdot 10 + 7 \cdot 24 = 240 + 168 = 408$). The foot of the perpendicular from the origin to the line $24x + 7y = 408$ is $E = (408 / (24^2 + 7^2)) \cdot (24, 7) = (408/625)(24, 7) = (9792/625, 2856/625)$.

F = foot from C on line AB . Line AB is $12x - 5y = 0$. The foot from $C = (17,0)$ is $F = C - ((12 \cdot 17 - 5 \cdot 0)/(12^2 + 5^2)) (12, -5) = (17, 0) - (204/169)(12, -5) = (425/169, 1020/169)$.

2.3 The circumcircle Ω .

Write Ω : $x^2 + y^2 + dx + ey + f = 0$. Through $B(0,0)$: $f = 0$. Through $C(17,0)$: $289 + 17d = 0 \Rightarrow d = -17$. Through $A(10,24)$: $100 + 576 - 170 + 24e = 0 \Rightarrow 506 + 24e = 0 \Rightarrow e = -253/12$. Hence Ω : $x^2 + y^2 - 17x - (253/12)y = 0$. (1)

3

STEP 1: THE POINT $X = EF \cap BC$

Lemma (harmonic conjugate). X is the harmonic conjugate of D with respect to B and C ; equivalently $(B, C; D, X) = -1$.

Proof. Because $\angle BEC = \angle BFC = 90$ degrees, the points B, C, E, F lie on the circle ω with diameter BC . Consider this circle. Its chord BC and its chord EF meet at X . Its chord BF ($=$ line AB) and its chord CE ($=$ line CA) meet at A . Its chord BE and its chord CF meet at H (the two altitudes BE, CF cross at the orthocentre). Thus, for the complete quadrilateral inscribed in ω with vertices B, F, E, C , the three diagonal points are X ($= BC \cap EF$), A ($= BF \cap CE$) and H ($= BE \cap CF$). Consequently the line AH is the polar of X with respect to ω . The line AH is the A -altitude, which meets BC at D . Since X also lies on BC , the polar of X meets BC at D , and therefore D and X are harmonic conjugates with respect to the points B, C where BC cuts ω : $(B, C; D, X) = -1$. $\square$

Computation of X . On the x -axis use the coordinates $B = 0, C = 17, D = 10$. With $X = (x_0, 0)$, the relation $(B, C; D, X) = -1$ reads $(DB/DC) : (XB/XC) = -1$, i.e. $XB/XC = -(DB/DC)$. Here $DB = 10, DC = 7$ (directed $B \rightarrow C$), so $XB/XC = -10/7$, giving $x_0/(x_0 - 17) = -10/7 \Rightarrow 7x_0 = -10x_0 + 170 \Rightarrow 17x_0 = 170 \Rightarrow x_0 = 10 \dots$ (that is D); the harmonic conjugate is the OTHER solution of $XB/XC = 10/7$ taken externally: $x_0/(x_0 - 17) = 10/7 \Rightarrow 7x_0 = 10x_0 - 170 \Rightarrow 3x_0 = 170$, so

$$X = (170/3, 0). \quad (2)$$

(Direct check by intersecting line EF with $y = 0$ from the coordinates of E and F gives the same value $170/3$; this is carried out and confirmed.)

4

STEP 2: THE POINT $Q = (\text{line } AX) \cap \Omega$

The direction of AX is $X - A = (170/3 - 10, -24) = (140/3, -24)$, parallel to $(35, -18)$. Parametrise the line AX by $(x, y) = (10 + 35t, 24 - 18t)$, t in $\mathbb{R}$, $t = 0$ giving A .

Substitute into (1). Compute each piece: $x^2 + y^2 = (10+35t)^2 + (24-18t)^2 = (100 + 700t + 1225t^2) + (576 - 864t + 324t^2) = 676 - 164t + 1549t^2$, $-17x = -170 - 595t$, $-(253/12)y = -(253/12)(24 - 18t) = -506 + (759/2)t$.

Adding, the constant term is $676 - 170 - 506 = 0$ (this just re-expresses that A lies on Ω), and the equation (1) becomes $1549t^2 - (759/2)t = 0 \Rightarrow t(1549t - 759/2) = 0$. The root $t = 0$ is A ; the second intersection corresponds to $t = 759 / (2 * 1549) = 759/3098$. Therefore $Q_x = 10 + 35 * 759/3098 = (30980 + 26565)/3098 = 57545/3098$, $Q_y = 24 - 18 * 759/3098 = (74352 - 13662)/3098 = 60690/3098 = 30345/1549$, i.e.

$$Q = (57545/3098, 30345/1549). \quad (3)$$

(One checks directly that Q satisfies (1): substituting (3) into the left side of (1) yields exactly 0.)

STEP 3: THE POINT $P = HQ \cap (\text{circle DEF})$ IS THE MID-POINT OF HQ

The circle through D, E, F is the nine-point circle of triangle ABC. It is the image of the circumcircle Ω under the homothety h centred at the orthocentre H with ratio $1/2$. (Standard fact: h maps the circumcentre O to the nine-point centre $N = (O + H)/2$, the midpoint of OH, and scales the radius by $1/2$; under h each vertex of the triangle maps to the Euler point, and the whole circumcircle maps onto the nine-point circle, which passes through D, E, F.)

Since Q lies on Ω , its image $h(Q) = H + (1/2)(Q - H) = (H + Q)/2$ lies on the nine-point circle, i.e. on circle DEF. Moreover $(H + Q)/2$ is the midpoint of HQ, which lies in the interior of the segment HQ. The line HQ meets the circle DEF in at most two points; the midpoint is one of them and is the one lying on the segment HQ (the second intersection of line HQ with the circle is the point $H + (1/2 - \delta) \dots$ in fact it lies strictly outside the segment, as a direct solve of the quadratic confirms: parametrising HQ by $H + u(Q - H)$, the two intersections occur at $u = 1/2$ and at a value $u < 0$). Hence the point P named in the problem is

$$P = (H + Q)/2. \quad (4)$$

Using (3) and $H = (10, 35/12)$: $P_x = (10 + 57545/3098)/2 = (88525/3098)/2 = 88525/6196$, $P_y = (35/12 + 30345/1549)/2$. For P_y put both terms over 18588 = $12 \cdot 1549$: $35/12 = 54215/18588$, $30345/1549 = 364140/18588$, sum = $418355/18588$, half = $418355/37176$. So $P = (88525/6196, 418355/37176)$. (5)

STEP 4: THE LENGTHS PQ AND PE

6.1 Length PQ.

By (4), $PQ = (1/2) HQ$. Now $Q - H = (57545/3098 - 10, 30345/1549 - 35/12) = (26565/3098, 309925/18588)$. Put both coordinates over the common denominator 18588 = $12 \cdot 1549$ (note $26565/3098 = 159390/18588$): $Q - H = (1/18588) (159390, 309925)$. Hence $HQ^2 = (159390^2 + 309925^2) / 18588^2$.

Numerator: $159390^2 + 309925^2 = 121,458,677,725$. This factors as $121,458,677,725 = 1549 \cdot 8855^2$, ($8855^2 = 78,411,025$), which one verifies by multiplication: $1549 \cdot 78,411,025 = 121,458,677,725$. Denominator: $18588^2 = 144 \cdot 1549^2$. Therefore $HQ^2 = (1549 \cdot 8855^2) / (144 \cdot 1549^2) = 8855^2 / (144 \cdot 1549)$, and, since $144 \cdot 1549 = (12)^2 \cdot 1549$, $HQ = 8855 / (12 \cdot \sqrt{1549})$, $PQ = HQ/2 = 8855 / (24 \cdot \sqrt{1549})$. (6)

6.2 Length PE.

From (5) and $E = (9792/625, 2856/625)$ put both points over the common denominator 23,235,000 = $625 \cdot 37176$ (here $37176 = 24 \cdot 1549$): $E = (364,027,392 / 23235000, 106,174,656 / 23235000)$, $P = (331,968,750 / 23235000, 261,471,875 / 23235000)$. Hence $E - P = (32,058,642 / 23235000, -155,297,219 / 23235000)$, $PE^2 = (32,058,642^2 + 155,297,219^2) / 23,235,000^2$.

Numerator: $32,058,642^2 + 155,297,219^2 = 25,144,982,756,018,125$. This factors as $25,144,982,756,018,125 = 625 \cdot 1549 \cdot 161161^2$, ($161161^2 = 25,972,867,921$), verified by multiplication: 161161^2

$= 25,972,867,921, 25,972,867,921 * 1549 = 40,231,972,409,629, 40,231,972,409,629 * 625 = 25,144,982,756,018,125$. Denominator: $23,235,000^2 = (3 * 5 * 1549 * 1000)^2 = 2^6 * 3^2 * 5^8 * 1549^2$. Therefore $PE^2 = (625 * 1549 * 161161^2) / (2^6 * 3^2 * 5^8 * 1549^2) = 161161^2 / (2^6 * 3^2 * 5^4 * 1549) = 161161^2 / (360000 * 1549)$, and, since $360000 = 600^2$, $PE = 161161 / (600 * \sqrt{1549})$. (7)

7 ===== STEP 5: THE RATIO PQ/PE

Dividing (6) by (7), the common factor $\sqrt{1549}$ cancels:

$$\frac{PQ}{PE} = \frac{8855 / 24}{161161 / 600} = \frac{8855 * 600}{24 * 161161} = \frac{8855 * 25}{161161}.$$

Now use the prime factorisations $8855 = 5 * 7 * 11 * 23$, so $8855 * 25 = 5^3 * 7 * 11 * 23 = 221375$, $161161 = 7^2 * 11 * 13 * 23$. Their greatest common divisor is $7 * 11 * 23 = 1771$, and $221375 / 1771 = 125$, $161161 / 1771 = 91$. Hence

$$PQ/PE = 221375 / 161161 = 125 / 91.$$

Finally $125 = 5^3$ and $91 = 7 * 13$ share no common factor, so $125/91$ is already in lowest terms. (Numerically $125/91 = 1.37362\dots$, matching the direct numerical evaluation of PQ/PE .)

8 ===== CONCLUSION

$$PQ / PE = 125 / 91, \quad \text{with } a = 125, b = 91 \text{ coprime.}$$

$$a + b = 125 + 91 = 216.$$

9 ===== REMARK (a structural check)

A coordinate-free reason why Q is special: by the power of the point X, $XA * XQ = XB * XC$ (X, A, Q collinear on Omega; X, B, C collinear), and also $XB * XC = XE * XF$, the power of X with respect to the circle omega on diameter BC (which contains B, C, E, F). Hence $XA * XQ = XE * XF$, so A, E, F, Q are concyclic. But A, E, F, H already lie on the circle with diameter AH (since angle AEH = angle AFH = 90 degrees). Therefore Q lies on the circle with diameter AH, giving angle AQH = 90 degrees, i.e. HQ is perpendicular to AX. (This perpendicularity is confirmed by the direction vectors: $HQ \sim (26565/3098, 309925/18588)$ is orthogonal to $AX \sim (35, -18)$.) This is the geometric content behind the clean factorisations appearing above; it is not needed for the final arithmetic, which is established rigorously in Steps 1-5.